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TASK 1A

International shipping plays a crucial role in today's global economy. Every day, individuals and businesses send cargo across borders. Air travel is the fastest and most efficient form of transport, capable of reaching almost any destination quickly. Due to its speed and reliability, air freight has become the preferred option for many.

However, air freight presents several challenges for consumers and businesses, including high shipping costs, limited cargo options, delivery delays, capacity restrictions, and poor visibility into order details. Mistakes during order entry can lead to errors, confusion, and financial loss. These issues often cause frustration and make planning difficult, especially when users lack control over shipment details or cannot verify important information.

That's why I created a Delphi program named: Jean's Express. The program gives consumers full control over cargo type, weight, fragility, and destination, enabling smarter and more affordable shipping decisions. It clearly displays costs, flight details, and cargo options before payment, helping users avoid unnecessary fees. Real-time validation prevents errors, while filtered flight matching ensures fast, reliable deliveries. This improves efficiency and planning, reducing frustration caused by delays and high prices.

My program has limitations, such as a limited selection of cities and cargo types. Customers cannot correct order mistakes themselves, and the program currently lacks real-time tracking updates. In the future, I plan to expand the number of cities, cargo types, and introduce customer support features.

TASK 1B

Role	Activity	Limitations
Consumer	The consumer will enter personal details and order information to place a shipment. This booking will be added as a new record in the database.	The user cannot fix errors or cancel an order once it has been placed. They will also not have access to administrative sections of the program, such as editing, deleting, or viewing all orders.
Consumer	The user can verify their order status by entering details. This means they will see their order information retrieved from the database.	Not secure or private, as the information can be viewed by any person if they have the required information about the order.
Casual visitor	They can view frequently asked questions and a brief company history and About Us section, which provides useful guidance and background information about the service.	The FAQ and About section may not be updated frequently, offer generalized questions, and lack personalization for individual customer issues.
Admin	The administrator will be able to view the entire database and have full control to manage shipping records. This includes deleting records, editing the flight's status, and viewing the overall highest or average cost of orders, or viewing orders that have been placed today.	No Authorization, as only a password is used for access.

TASK 2

Databases in Delphi are used to efficiently store, manage, and retrieve large amounts of data. For my flight service, the database will store information about flights and customer orders. The admin will be able to delete, edit and search for records to create data and improve overall efficiency.

TblCargo

It will store all information about orders that the costumers place.

Field Name	Data Type	Field Size
Cargo_ID	AutoNumber	Integer
CargoType	Short Text	15
Weight (kg)	Number	Integer
Cost	Currency	-
Fragile	Yes/No	-
Shipper	Short Text	15
Receiver	Short Text	15
Flight_ID	Short Text	4

Example:

Cargo_ID	CargoType	Weight (kg)	Cost	Fragile	Shipper	Receiver	Flight_ID
1	Machinery	34	R1 887,84	☒	Sam	Jerry	FL1
2	Electronics	432	R7 887,39	☒	Dave	Allen	FL6
3	Dangerous Goods	44	R1 578,72	☒	William	John	FL14
4	Pharmaceuticals	445	R8 265,63	☐	Adam	Jeff	FL17
5	Art	34	R1 941,66	☒	Charles	Kayla	FL11
6	Machinery	220	R4 623,00	☐	Quinton	Xavier	FL8
7	Machinery	56	R1 600,80	☐	Mary	George	FL20
8	Electronics	334	R7 443,03	☒	Wilton	Ben	FL12
9	Dangerous Goods	500	R7 969,50	☐	Nathan	Todd	FL9
10	Pharmaceuticals	332	R6 762,00	☒	Billy	Zack	FL14

TblFlights

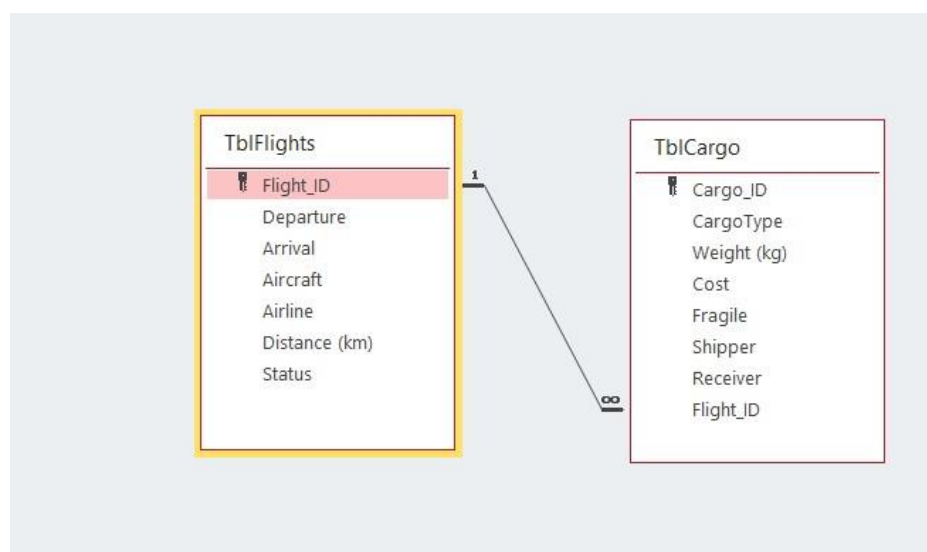
Contains the information regarding the flights that will ship the package.

Field Name	Data Type	Field Size
Flight_ID	Short Text	4
Departure	Short Text	15
Arrival	Short Text	15
Aircraft	Short Text	15
Airline	Short Text	15
Distance (km)	Number	Integer
Status	Short Text	10

Example:

Flight_ID	Status	Departure	Arrival	Aircraft	Airline	Distance (km)
FL1	Delivered	Cape Town	Dubai	Boeing 777	DHL Aviaton	8000
FL2	In Transit	Cape Town	Hong Kong	Boeing 767	Cargolux	13000
FL3	In Transit	Cape Town	London	Airbus A330	DHL Aviaton	10000
FL4	Delivered	Cape Town	New York	Douglas DC-10F	FedEX Express	13000
FL5	Delivered	Dubai	Cape Town	Airbus A330	Emirates SkyCa	8000
FL6	Delivered	Dubai	Hong Kong	Boeing 767	Emirates SkyCa	6500
FL7	Delivered	Dubai	London	Douglas DC-10F	UPS Airlines	6000
FL8	Booked	Dubai	New York	Boeing 747	Emirates SkyCa	11500
FL9	Delivered	Hong Kong	Cape Town	Boeing 777	FedEX Express	13000
FL10	Booked	Hong Kong	Dubai	Boeing 767	Emirates SkyCa	6500

Relationship



Each flight can have multiple packages, which means I'm using a 'One-To-Many' relationship.

TASK 3A

Where Objects Add Value

Classes and objects organize code, improve reusability, and simplify maintenance by structuring data and functionality in Delphi.

Rather than a large set of IF THEN ELSE statements, each flight is its own TFlight object carrying its departure, arrival, code and distance. Adding, removing or modifying routes becomes easier because you are changing data in one place, not hunting through dozens of IF statements and code. This will save space and time for the admin as there are fewer lines of code used that serves the same purpose.

How I plan to use it

Using classes like “TFlight” cleanly organizes flight data, allowing for structured storage, easier manipulation, and scalability. Constructors simplify initialization, while procedures like “LoadFlights” centralize setup, ensuring efficiency and clarity.

The “FindFlight” procedure will run through “arrFlights” to verify whether a flight with specific departure and arrival cities exists. If found, it retrieves the flight code and distance, ensuring accurate, validated data before processing—preventing errors and improving reliability.

TFlight = Class	
- sDeparture: string - sArrival: string - sFlight: string - iDistance: Integer	PRIVATE PUBLIC
+Create(	sDeparture: string; // City where the flight departs sArrival: string; // City where the flight arrives sFlight: string; // Unique flight ID/code iDistance: Integer; // Distance between destinations)
+LoadFlights()	
+FindFlight	(FlightID : String) : Boolean
-	

Uses



Method	Type	Return	Parameters	Purpose	Scope
LoadFlights	Procedure	-	None	Populate the program's internal array of flights with constant information.	Can't load from a file or database.
TFlight.Create	Constructor	-	Departure, Arrival, Flight: string; Distance: Integer	Initialize a single flight's data before adding the record to the database.	You need to manually add each flight to arrFlights—might be prone to mistakes if forgotten.
FindFlight	Function	Boolean	sFlightID: String;	Confirm that the flight exists, ensuring that data is correct and error-free.	Doesn't search the actual arrFlights array, so if the flight list changes, the function is inaccurate.

Task 3B

File	Use	Purpose
About_Us.txt	Input	Contains information regarding commonly asked questions and a brief history of the Company.
Orders.txt	Input/Output	Store, update, and display orders made during the day, allowing one program to write new orders (Consumer) and another to read them. (Admin)

Example of “About Us” File in Delphi:

About Us

[? Help](#)

FAQ

✓ About Us

i

Jean's International Shipping has been shipping cargo reliably and efficiently since 1998. It is our number one priority to ensure that cargo is transported safely and swiftly. In our almost 30 years of service, we have successfully delivered over 120000 packages. Expect friendly staff who are determined to assist you and competitive prices that are hard to beat. Thanks to our latest software update, you can now track the status of your packages in real-time, with instant updates and no delays. At our company, we're committed to investing in the latest advancements in logistics technology and aviation. Our aircraft feature cutting-edge systems to ensure seamless, optimized, and secure transportation. We pledge to be eco-friendly and we regularly donate to charities and other organizations. Our service has built a reputation for making users feel confident and satisfied. Why wait? Book a shipment today and experience our award-winning service!

Close

Example of Data from Orders.txt:

```
2025/04/04,20,Electronics,67,R2 605,32,False,Gernard,Dan,FL2
2025/04/04,21,Machinery,22,R1 200,60,False,Bill,Jack,FL6
2025/04/04,22,Dangerous Goods,333,R4 971,45,False,Frank,Carl,FL7
```


TASK 3C

Array	Data Type	Size	Type	Purpose
ArrFlights	TFlight (Class, scroll up to see variables used)	[0..19]	Standard	Contains all the data about every Flight. It will be used to determine which Flight the consumer will use.
ArrOrders	String	[1..200, 1..8]	2 Dimensional	Contains all records, fields and data of orders that were placed today. It will also be used to display the orders.

Example of ArrFlights:

1	2	3	4
Cape Town	Dubai	FL1	8000

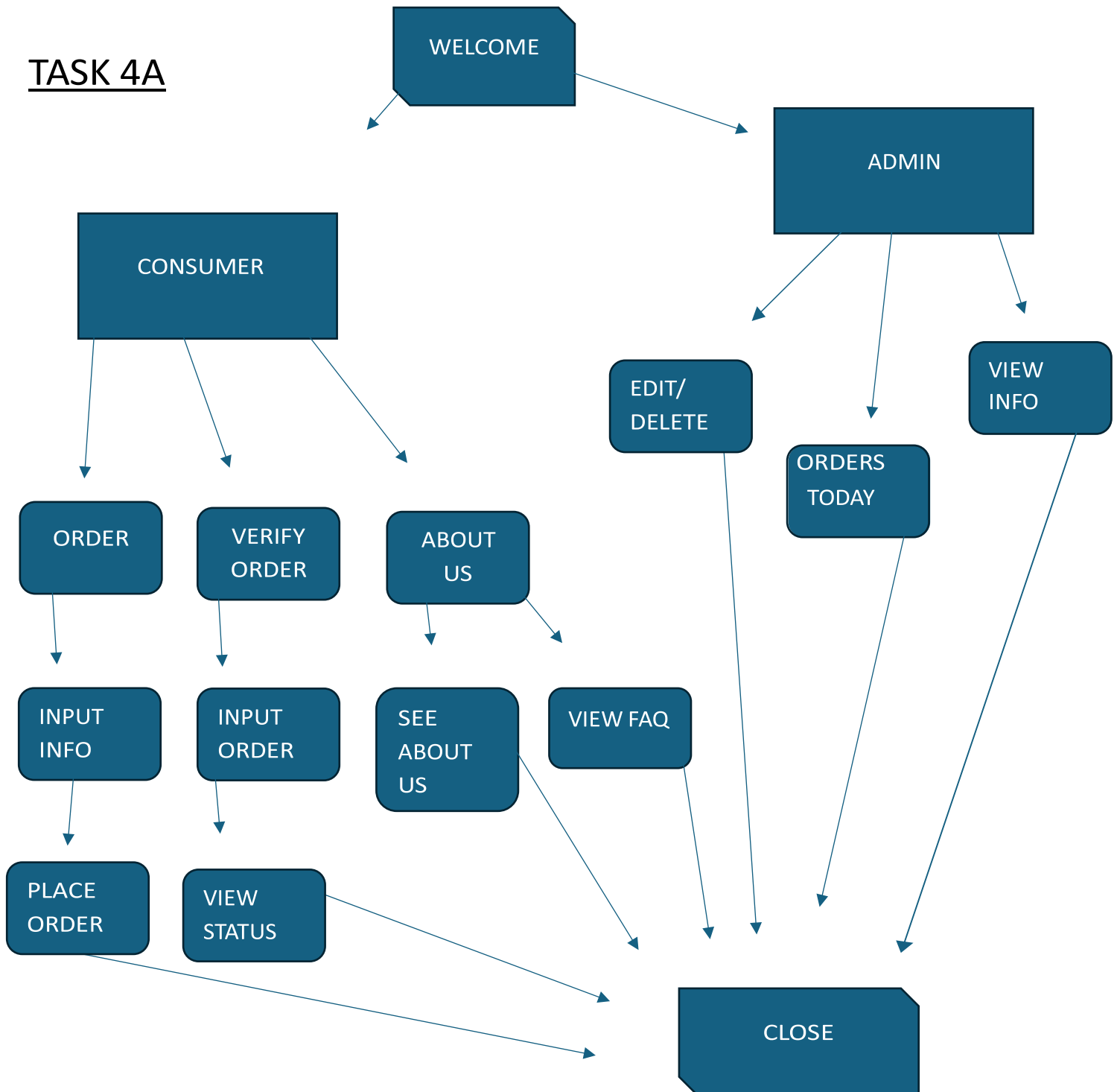
```
arrFlights[0] := TFlight.Create('Cape Town', 'Dubai', 'FL1', 8000);
```

Example of ArrOrders (2 Dimensional) :

	1	2	3	4	5	6	7	8
1	20	Electronics	67	R2605,32	False	Gerald	Dan	FL2
2	21	Machinery	82	R1200,89	False	Bill	Jack	FL6

```
2025/04/04,20,Electronics,67,R2 605,32,False,Gernard,Dan,FL2  
2025/04/04,21,Machinery,22,R1 200,60,False,Bill,Jack,FL6
```

TASK 4A



TASK 4B

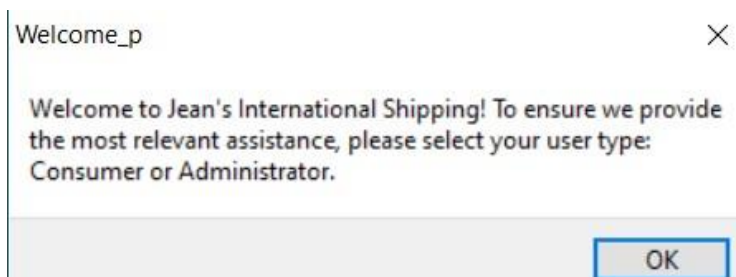
I will use a clean, simple design with clear buttons, readable fonts, and logical layouts, to ensure that users can easily navigate my Delphi Program and complete tasks without confusion or frustration. I will use messages in my code to detect incorrect input and display clear, helpful error messages to guide users to enter valid information and avoid errors. A help button will display instructions or tips in a pop-up, guiding users through tasks and answering common questions.

Welcome Form

This is the welcome page. The user will indicate what type of user he/she is.



Example of Help Message (Page 3) :



Component	Name	Purpose
Form	FrmWelcome	-
Image	Img1	Create a GIF to make a prettier GUI
Image	Img2	Used for Background
Panel	Pnl1	-
Panel	Pnl2	-
BitButton	BitBtnConsumer	Open Consumer Form
BitButton	BitBtnConsumer	Open Admin Form
BitButton	BitBtnHelp	Show Help Message

Page 1: Consumer

The consumer will provide his information and place an order.

ORDER HERE

? Help

Cargo Type

Weight (kg)

0

Departure

Arrival

Shipper

Receiver

☐ Fragile Shipment

✓ Calculate

Close

The user will input his details, and will then click 'Calculate to view his order's information.

Example of Output:

✓ Calculate

Confirm

Reset

Close

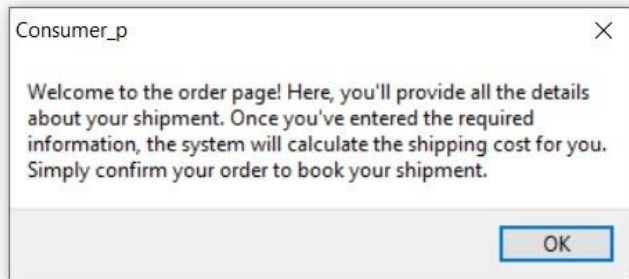
Cargo Type	Weight	Departure	Arrival	Shipper	Receiver	Fragile	Cost	FlightID
Dangerous Goods	22kg	Cape Town	London	Jean	Sam	True	R 1 851,96	FL3

If your order is correct, Confirm it to Book your Shipment.

If confirmed, the order will be placed and displayed in the form of a dbgrid.

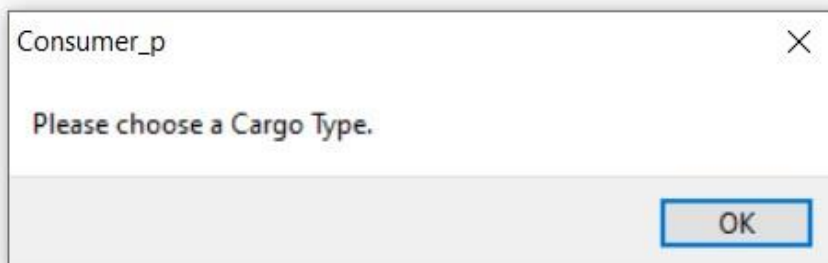
Cargo_ID	CargoType	Weight (kg)	Cost	Fragile	Shipper	Receiver	Flight_ID	
▶ 68	Dangerous Goods	50	2277	True	Sam	Tristan	FL11	^ v

Example of Help Message:



Example of Validate Message:

(If Cargo Type is not selected)



Component	Name	Purpose
Form	FrmConsumer	-
Label	LblCargo	-
Label	LblWeight	-
Label	LblDeparture	-
Label	LblArrival	-
Label	LblShipper	-
Label	LblReceiver	-
BitButton	BitButtonCalculate	Calculate the Cost and display Order
BitButton	BitButtonHelp	Show Help Message
BitButton	BitButtonClose	Close Form
BitButton	BitButtonConfirm	Confirm Order
Richedit	BitButtonReset	Display Order Details
ComboBox	CmbbxDeparture	-
ComboBox	CmbbxArrival	-
Checkbox	ChckbxFragile	-
Spinedit	SedtWeight	-
Edit	EdtShipper	-
Edit	EdtReceiver	-
Panel	Pnl1	-

Page 2: Admin

This is where the admin will have full access to the database to see orders and improve the service.

The screenshot shows the 'ADMINISTRATOR' interface. At the top, there's a header with the title 'ADMINISTRATOR', a 'Help' button, and a 'Close' button. Below the header is a table with the following data:

Cargo_ID	CargoType	Weight (kg)	Cost	Fragile	Shipper	Receiver	Flight_ID
1	Machinery	34	1887,84	True	Sam	Jerry	FL1
2	Electronics	432	7887,39	True	Dave	Allen	FL6
3	Dangerous Goods	44	1578,72	True	William	John	FL14
4	Pharmaceuticals	445	8265,625	False	Adam	Jeff	FL17
5	Art	34	1941,66	True	Charles	Kayla	FL11

Below the table, there are several buttons: 'Sort Names Alphabetically', 'Highest Cost', 'Average Cost', 'Select Cargo_ID', 'Delete Record', 'Edit Flight Status', and 'Today'.

Each button serves a different purpose, so it does not matter in which order they are clicked.

Example of Output:

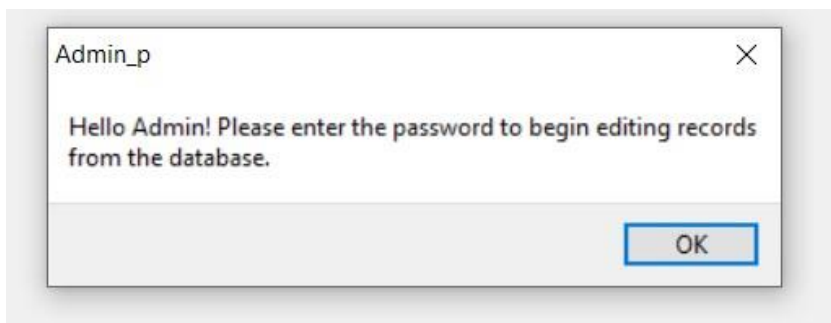
If Admin wants to see the highest cost of any order.

The screenshot shows the 'ADMINISTRATOR' interface with the 'Highest Cost' button highlighted. The table below the header shows a single record:

Cargo_ID	CargoType	Weight (kg)	Cost	Fragile	Shipper	Receiver	Flight_ID
17	Dangerous Goods	500	9335,7	True	Gerald	Devin	FL8

The 'Highest Cost' button is highlighted with a blue border, indicating it is the selected sorting option.

Example of Help Message:



Example of Validate Message:

(If package does not exist)



Component	Name	Purpose
Label	LblAdmin	-
Panel	Pnl1	-
Dbgrid	DbGridCargo	Display the records being edited.
Label	LblPassword	
Edit	EdtPassword	Where User types the password
BitButton	BitBtnHelp	Show Help Message
BitButton	BitBtnClose	Close Form
BitButton	BitBtnSort	Sort Alphabetically
BitButton	BitBtnHighest	Display Highest Cost
BitButton	BitBtnAverage	Calculate Average Cost
BitButton	BitBtnSelect	Select One Record
BitButton	BitBtnEdit	Edit Record
BitButton	BitBtnDelete	Delete Record
BitButton	BitBtnToday	Display Orders Placed Today

TASK 5

DATA INPUT					
Name	Source	Component	Type	Format	Example
sShipper	Keyboard	Edit	String	-	Jean
bFragile	Mouse	Checkbox	Boolean	True/False	True
iWeight	Mouse/Keyboard	Spinedit	Integer	-	50
sCargoID	Keyboard	InputBox	String	-	8
sDateFile	File	Orders.txt	String	YYYY/MM/DD	2025/04/05

INPUT VALIDATION				
Name	Data Type	Component	Value	Message
sCargo	String	ComboBox	EMPTY	Please select the Type of Cargo.
sFAQ	String	Combobox	EMPTY	Please select a question.
sReceiver	String	Edit	EMPTY	Please type in the Name of the Receiver.
sFlightID	String	Edit	EMPTY OR >20	Please select another FlightID.
iWeight	Integer	Spinedit	0 OR > 500	Please input an appropriate weight for the package.
FindFlight	Boolean	(Function)	False	Please choose an appropriate Departure and Arrival City.
rUserCost	Real	Edit	NOT rTotalCost	Please type in the correct cost of the Order.

DATA PROCESSING

Name	Type	Source	Component	Example
sCargo	String	CmbbxCargo	Combobox	Electronics
sDeparture	String	CmbbxDeparture	Combobox	Dubai
sArrival	String	CmbbxArrival	Combobox	Cape Town
sUserCargo	String	SedtCargoID	Edit	12
sUserFlight	String	EdtFlightID	Edit	FL4
iWeight	Integer	SedtWeight	Spinedit	250
iCargonum	Integer	TblCargo	AdoTable	87
rUserCost	Real	EdtUserCost	Edit	2100,86
bFragile	Boolean	ChkbxFragile	Checkbox	False
sDate	String	Date	-	2025/04/01

Name	Process	Type	Example
rCost	$iDistance * 0.1 + iWeight * 10$	Real	4301,25
rCargo	If scargotype = 'Art' then rCargo * 1.05	Real	4516,31
rFragile	If bFragile = true then rCost * 1.2	Real	5419,58
rTax	$rcost * 0.15$	Real	812,95
rTotalCost	$rCost + rTax$	Real	6232,51
rHighestCost	<pre>while not tblcargo.Eof do begin rcost := tblcargo['Cost']; if rcost > rhighestcost then begin rhighestcost := rcost; end; tblcargo.Next; end;</pre>	Real	9832,84
rAverage	<pre>while not tblcargo.Eof do begin rsum := rsum + rcost; tblcargo.Next; end; ravg := rsum / itblcount;</pre>	Real	3892,48

DATA OUTPUT

Data	Type	Format	Component	Output	Example
sReceiver	String	-	Richedit	RedtOrder	Hank
rTotalCost	Real	Currency	Richedit	RedtOrder	R7629,91
bFragile	Boolean	True/False	Richedit	RedtOrder	True
iWeight	Integer	-	Richedit	RedtOrder	45 kg
rAverage	Real	Currency	Label	LblAverage	R5484,29
sDate	String	YYYY/MM/DD	Stringgrid	Strgrd	2025/04/05